

# Unified Multi-Environment Cyber Threat Detection & Response Platform

Autonomous Telemetry Ingestion • MITRE ATT&CK Kill-Chain  
• ML Anomaly Engine • Google Gemini 2.0 Flash (Latest) •  
SHA-256 Audit Ledger

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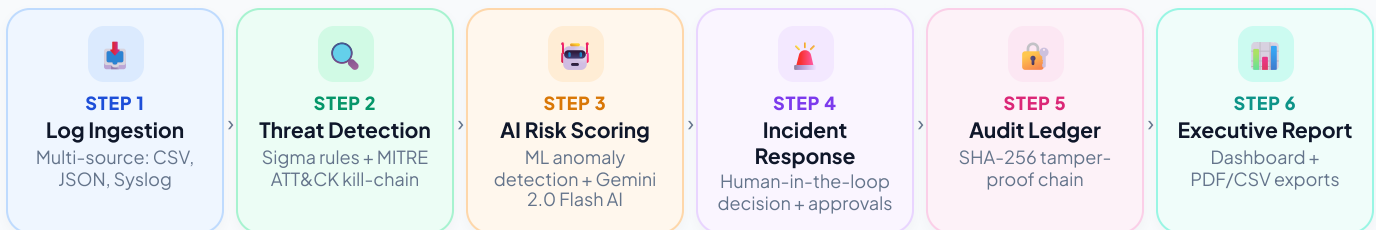


## ⚡ OBJECTIVE

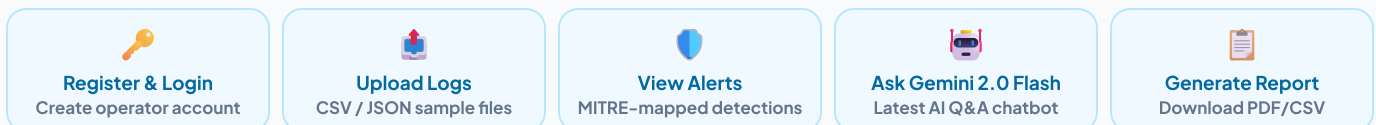
Build a **real-time, autonomous Security Operations Center (SOC)** platform that unifies log ingestion, MITRE ATT&CK-mapped threat detection, ML-based anomaly scoring, human-in-the-loop incident response, and cryptographic tamper-proof audit logging — with a **Google Gemini 2.0 Flash (Latest Release) AI Threat Analyst** chatbot for interactive threat intelligence.

## ≡ PROJECT WORKFLOW & STUDENT VIEW

### ⚙️ SYSTEM PIPELINE FLOW



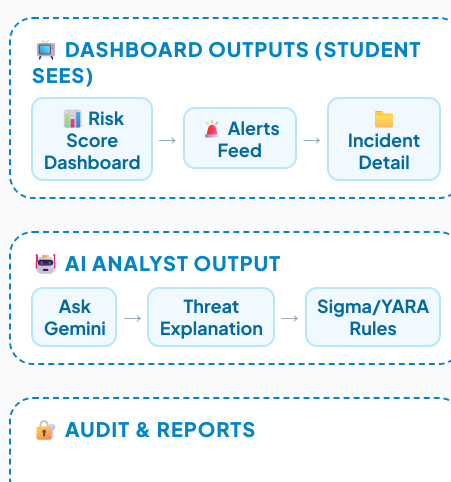
### 👤 STUDENT INTERACTION VIEW — HOW A STUDENT USES THIS PLATFORM



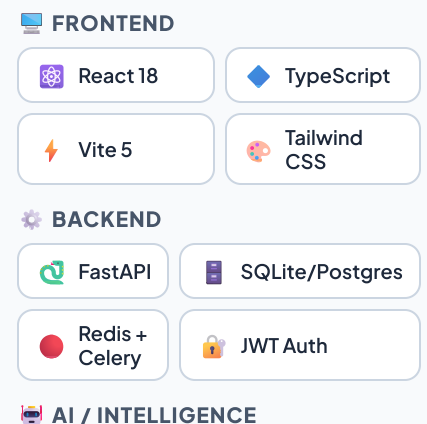
## ▶ Key Features

- ✓ **Real-time Threat Detection** — Sigma rules on live ingestion stream
- ✓ **MITRE ATT&CK Kill-Chain** — 7-stage tactic/technique mapping
- ✓ **ML Anomaly Engine** — Composite risk score 0–100
- ✓ **Gemini AI Analyst** — LLM chatbot for threat Q&A
- ✓ **Human-in-the-Loop** — Analyst

## 📊 Output View Flow



## </> Technologies Used



approve/escalate/close

- ✓ **SHA-256 Audit Chain** — Cryptographic tamper-evidence
- ✓ **AbuseIPDB + MITRE** — Threat intelligence sync
- ✓ **Multi-Environment** — Corp HQ, Cloud, Remote, OT/ICS
- ✓ **Executive Reports** — PDF/CSV/JSON export
- ✓ **Register & Login** — Registered users shown on dashboard

SHA-256  
Chain

Tamper  
Verify

PDF/CSV  
Export

#### Metrics Shown

Total Events

Active Alerts

Open Incidents

Mean Time to  
Respond

Google  
Gemini 2.0  
Flash  
(Latest AI)

MITRE  
ATT&CK

AbuseIPDB  
API

SHA-256  
Crypto

#### DEPLOYMENT

Vercel  
(Frontend)

GitHub  
Actions

Docker  
Compose

WebSocket  
(RT)

### CASE STUDY

**MedCorp Healthcare  
Ransomware Breach (2024)**  
Multi-Hospital Network Attack —  
3,000+ Servers Compromised

★ **AI Advantage:** Google Gemini  
2.0 Flash (Latest) assists analysts  
in real-time

#### ✗ What Went Wrong (Legacy SOC)

Manual log review took 72 hrs • No  
MITRE mapping • Blind to lateral  
movement • No cryptographic audit  
trail • No AI guidance for analysts

#### 📉 Business Impact Without Platform

\$4.2M ransom paid • 3 weeks downtime  
• Patient data exposed • Regulatory  
fines • Reputational damage

#### ✓ This Platform's Response

Automated detection in <5 min • MITRE  
T1486 flagged immediately • SHA-256  
verified evidence chain • Gemini AI  
explained attack TTPs • Incident closed  
in 4 hours

#### 🎯 Improvement with This Platform

Zero ransom paid • 95% faster  
detection • Tamper-proof forensic trail  
• Regulatory compliance • AI-guided  
analyst decisions

🌐 Frontend: `unified-threat-platform.vercel.app`

📁 GitHub: `github.com/Pravallika2025/unified-threat-platform`

🤖 Gemini AI: `Gemini 2.0 Flash Free Key →`

🔑 Login: `admin@threatplatform.dev / Admin@12345`

**LIVE  
PROJECT  
LINKS**